

# Analysis of Variance (ANOVA)

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# Introduction

- ▶ The “halo effect” is a hypothesized cognitive bias where a positive impression of one aspect of a person/brand leads to other aspects of that same person/brand being viewed more favorably than they should
- ▶ Today we'll look at data from the article: “Beauty is Talent: Task Evaluation as a Function of the Performer's Physical Attraction” published in *The Journal of Personality and Social Psychology* in 1974
  - ▶ 60 undergraduate males scored (from 0 to 25) an essay supposedly written by a female undergraduate
  - ▶ Attached to each essay was a photo of the supposed author that was randomly assigned from one of the following conditions: “attractive”, “unattractive”, or “none”

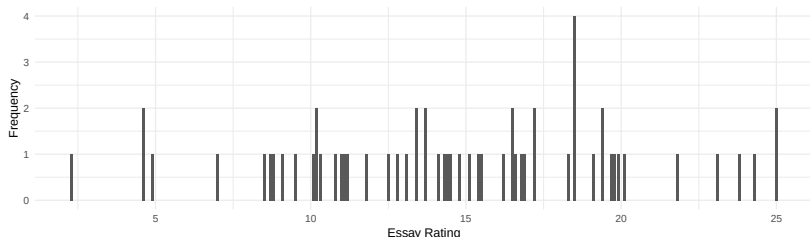
# Hypothesis testing

There are two types of hypotheses we might consider for this experiment:

1. **global hypothesis** - Is an essay's rating associated with the type of photo attached to it?
2. **pairwise" hypotheses** - Do ratings in a particular condition (ie: "attractive") differ from another condition (ie: "none")?
  - ▶ There are 3 different pairwise hypotheses in this example
  - ▶ The pairwise hypotheses can be evaluated using  $t$ -tests. However, type I errors are a concern
  - ▶ Analysis of Variance (ANOVA) allows us to evaluate the global hypothesis with a single test

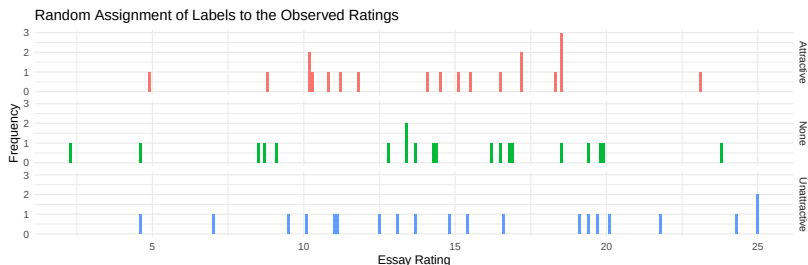
# The Null Hypothesis for ANOVA

If the experimental condition and essay rating are *independent*, we'd expect the ratings in each condition to follow the same distribution. Below is the overall distribution of scores:



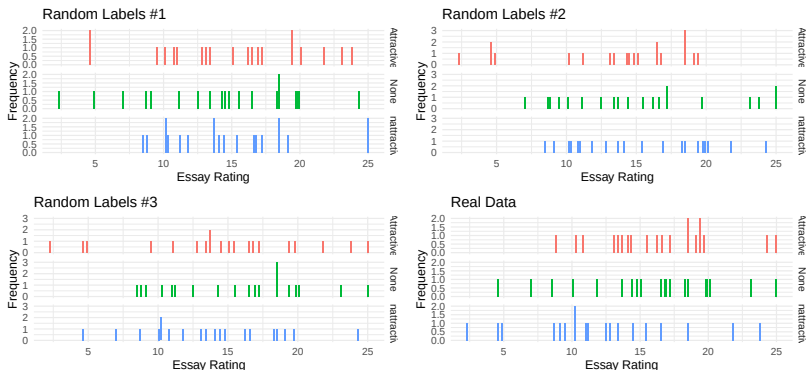
# The Null Hypothesis for ANOVA

Under ANOVA's null hypothesis of *no association*, we'd expect ratings in group to be sampled from the overall distribution. Below we simulate this by randomly giving each data-point a group label (unrelated to its actual group):



# Hypothesis Testing

ANOVA aims to determine whether the observed distribution *within groups* significantly deviates from what we'd expect if group labels were random (ie: no association between “group” and “rating”):



# Predictions

- ▶ To compare the observed and expected distributions, ANOVA looks at *predicted values*
  - ▶  $\hat{y}_i$  is a prediction of  $y_i$ , the  $i^{th}$  data-point's outcome (rating)
- ▶ Predictions can be made assuming  $H_0$ , or using relationships in the observed data

| Condition (group) | Mean Rating | Standard Deviation | n  |
|-------------------|-------------|--------------------|----|
| Attractive        | 16.4        | 4.3                | 20 |
| None              | 15.6        | 5.2                | 20 |
| Unattractive      | 12.1        | 5.4                | 20 |
| Overall           | 14.7        | 5.3                | 60 |

**Questions:** For a data-point in the “unattractive photo” group, what is its predicted rating under  $H_0$ ? What would you predict it's rating if you used the relationships present in the real data?

# Predictions and Models

In ANOVA the null hypothesis reflects a *statistical model*:

$$y_i = \mu + \epsilon_i$$

- ▶  $y_i$  is the  $i^{th}$  data-point's observed outcome,  $\mu$  is an overall mean, and  $\epsilon$  is an error that is assumed to follow a  $N(0, \sigma)$  distribution
  - ▶ Because the errors have an expected value of zero, this model predicts  $\hat{y}_i = \mu$ , the overall mean, for every data-point (all  $i \in \{1, 2, \dots, n\}$ )
- ▶ This is known as the **null model**



# Predictions and Models

ANOVA also considers an **alternative model**:

$$y_i = \mu_i + \epsilon_i$$

- ▶ This model looks similar, but  $\mu_i$  is a group-specific mean that depends upon which group the  $i^{th}$  data-point belongs to
  - ▶ Thus, this model predicts a *group-specific* mean for data-point, thereby reflecting an association between “group” and “rating”

# Theoretical vs. Fitted Models

Like when we first learned about regression, it is important to note the distinction between the *theoretical* and *fitted* models in ANOVA:

|             | <b>Null Model</b>        | <b>Alternative Model</b>   |
|-------------|--------------------------|----------------------------|
| Theoretical | $y_i = \mu + \epsilon_i$ | $y_i = \mu_i + \epsilon_i$ |
| Fitted      | $\hat{y}_i = \bar{x}$    | $\hat{y}_i = \bar{x}_i$    |

*Note:*  $\bar{x}$  is calculated using the entire sample, but  $\bar{x}_i$  is the group-specific mean for the  $i^{th}$  data-point's group

**Questions:** How could you *summarize* how well each model fits the observed data?

# Summarizing a Model

Under the any model each subject deviates from their prediction by a **residual**:

$$\begin{aligned}r_i &= \hat{y}_i - y_i \text{ (Definition of a residual)} \\ &= \bar{y} - y_i \text{ (Residuals for the null model)}\end{aligned}$$

We can *summarize* the total variability of the null model's predictions using a **sum of squares**:

$$SST = \sum_i r_i^2 \text{ for the null model}$$

We call this *SST* (sum of squares total) because it is the *largest possible* sum of squares (of any justifiable model)

## Summarizing a Model

The alternative model can also be summarized using a **sum of squares**:

$$SSE = \sum_i r_i^2 \text{ for the alternative model}$$

where  $r_i = \bar{y}_i$  (Residuals for the alt model)

We call this *SSE* because it summarizes the unexplained variability (errors) of the model that uses “group”

# Creating a Test Statistic

- ▶ In ANOVA we ask: *“does the grouping variable improve model fit beyond what might be expected due to random chance?”*
  - ▶ This can be assessed using the **F-test statistic**:

$$F = \frac{(SST - SSE)/(d_1 - d_0)}{\text{Std. Error}}$$

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- ▶ The  $F$  statistic is the *standardized drop* in the sum of squares *per additional parameter* used by the alternative model

# Randomization F-tests

- ▶ We started off by considering what the data might look like if we randomly assigned group labels (slide 5/6)
  - ▶ If we did this many times, we could get an idea of how the  $F$ -statistic is distributed under the null hypothesis
  - ▶ This StatKey menu allows us to perform such a simulation

Here is a link to the data:

[https://remiller1450.github.io/data/halo\\_effect.csv](https://remiller1450.github.io/data/halo_effect.csv)



# The F-distribution

- ▶ Under the null hypothesis (ie: presuming the null model is true), this  $F$ -statistic follows an  $F$ -distribution that depends upon two different degrees of freedom ( $df$ ) parameters
  - ▶ The *numerator*  $df$  is  $d_1 - d_0$
  - ▶ The *denominator*  $df$  is  $n - d_1$
- ▶ We can use StatKey to view various  $F$ -distribution curves
  - ▶ The observed  $F$ -statistic is compared to the  $F$ -distribution to determine the  $p$ -value

# What is the Standard Error?

- ▶ We've seen that standard errors tend to look like a measure of variability divided by the sample size
- ▶ In the ANOVA setting:

$$\text{Std. Error} = \frac{SSE}{n - d_1}$$

- ▶ This is the sum of squares of the alternative model divided by its *degrees of freedom*,  $df = n - d_1$
- ▶ Using this standard error, the  $F$  statistic can be expressed:

$$F = \frac{(SST - SSE)/(d_1 - d_0)}{SSE/(n - d_1)}$$

## Simplifying the $F$ -statistic

Because this test statistic looks complex, statisticians define the “sum of square groups” as:  $S SG = SST - SSE$ , making the  $F$ -statistic:

$$F = \frac{SSG / (d_1 - d_0)}{SSE / (n - d_1)}$$

This is further simplified by denoting a sum of squares defined by its degrees of freedom as a **mean square**:

$$F = \frac{MSG}{MSE}$$

Here  $MSG$  is the mean square of “groups”,  $MSE$  is the mean square of “error”

# What Should You Know?

Calculating the  $F$ -statistic and the corresponding  $p$ -value are too tedious to perform by hand, but you are expected to be familiar with **ANOVA tables**, which are how software like R will report the results of an ANOVA test:

| Source    | $df$        | Sum Sq. | Mean Sq. | $F$ -statistic | $p$ -value               |
|-----------|-------------|---------|----------|----------------|--------------------------|
| "Group"   | $d_1 - d_0$ | $SSG$   | $MSG$    | $MSG/MSE$      | Use $F_{d_1-d_0, n-d_1}$ |
| Residuals | $n - d_1$   | $SSE$   | $MSE$    |                |                          |
| Total     | $n - d_0$   | $SST$   |          |                |                          |

You might be asked to fill in the missing components of such a table, interpret a printed table, or relate an ANOVA table to visualizations or models

# What Should You Know? (cont.)

In addition to understanding the components of ANOVA table output, you should have a high-level understanding of the  $F$ -test:

- ▶ The ANOVA  $F$ -test involves a *nominal categorical* explanatory variable and a *quantitative* response variable
- ▶ The null model states that a single, overall mean is sufficient (indicative of independence), while the alternative model uses group-specific means (indicative of an association)
  - ▶ The  $F$ -statistic compares the performance of these two models on the sample data using *sums of squares*
- ▶ Under the null hypothesis of independence, the  $F$ -statistic should follow an  $F$ -distribution, which is used to determine the  $p$ -value
  - ▶ A small  $p$ -value provides evidence of an association